Technical Report: Diagnostic Analysis of the National Health Service in England using Python

Introduction

The National Health Service (NHS) is a publicly funded healthcare system in England. The country has a growing population of over 60 million and the NHS must expand to match this increase. To inform appropriate strategy the NHS needs to answer four key business questions:

1. Has there been adequate staff and capacity in the networks?
2. What was the actual utilisation of resources?
3. What percentage of appointments were missed, and how can this be reduced?
4. Can social media be leveraged to indicate operational pressure?

We refined these questions into targeted queries, utilising internal and external datasets to generate insights and recommendations.

Analytical Approach

Datasets

1. Actual Duration (csv)

Overview: Dataset containing daily appointment information at the Integrated Care Board (ICB) level, including location, date, and duration.

Limitations: Duration data is categorical (1-5 minutes) rather than numeric, so reduces precision. Approximately 10-15% of appointments lacked duration information.

2. Appointments Regional (csv)

Overview: Dataset containing monthly aggregated information on appointments including ICB location, appointment month, status, mode, healthcare professional type, and time between booking the appointment and attending.

Limitations: Aggregated at ICB level, so does not provide opportunity to identify individual practices. Between 10-11% of inputs into the dataset include unknown values or values affected by data quality.

3. National Categories (csv)

Overview: Dataset containing daily appointment data across service settings, context types and national categories.

Limitations: Approximately 4% of entries contain unknown mapping or inconsistent mapping.

4. Tweets (xlsx)

Overview: Dataset containing tweets accompanied by healthcare related hashtags. Useful for understanding public sentiment around healthcare.

Limitations: Not specific to the NHS. Opinion-based qualitative data.

Data Cleaning, Validation and Standardisation

Before analysis, datasets were reviewed in Python using pandas. Missing values were identified using .isna() and dates were standardised using pd.to_datetime(). Datasets were filtered from August 2021 onwards and monthly aggregates created with .groupby(). Columns were renamed for consistency, and inputs such as ‘Unknown’ or ‘Unmapped’ were retained but flagged as recurring issues.

Exploratory Data Analysis (EDA)

EDA was conducted using pandas, matplotlib, and seaborn. This allowed structure assessment, anomaly detection anomalies and ensured consistency across the data sources. Functions used to ensure data integrity included *.describe()*, *.info()*. Visuals were generated and refined, rotating month labels and avoiding scientific notation.

Feature-level Analysis

Feature-level summaries examined total appointments, capacity utilisation with a benchmark, healthcare professional types, appointment attendance, appointment mode and time between bookings and appointments. Various charts were generated to visualise these summaries, including line charts, bar charts (grouped and stacked), and boxplots.

External Data

Tweets were provided following data scraping in an excel dataset of healthcare-related tweets. Hashtags were cleaned and standardised to lowercase using *.str.lower()*, counted for frequency and a frequently recurring hashtag (#healthcare) was excluded to generate a readable visual. Engagement metrics were examined contextually, however generated a debate around sentiment versus amplification bias.

Visualisation and Insights

For accessibility, the Tab10 and Tab20 colour palette were used to ensure charts are readable for stakeholders with colour vision deficiencies.

Business Questions

1. *Has there been adequate staff and capacity in the networks?*

Measure: Average daily appointments by month. Included is an NHS provided benchmark capacity limit of 1.2 million.

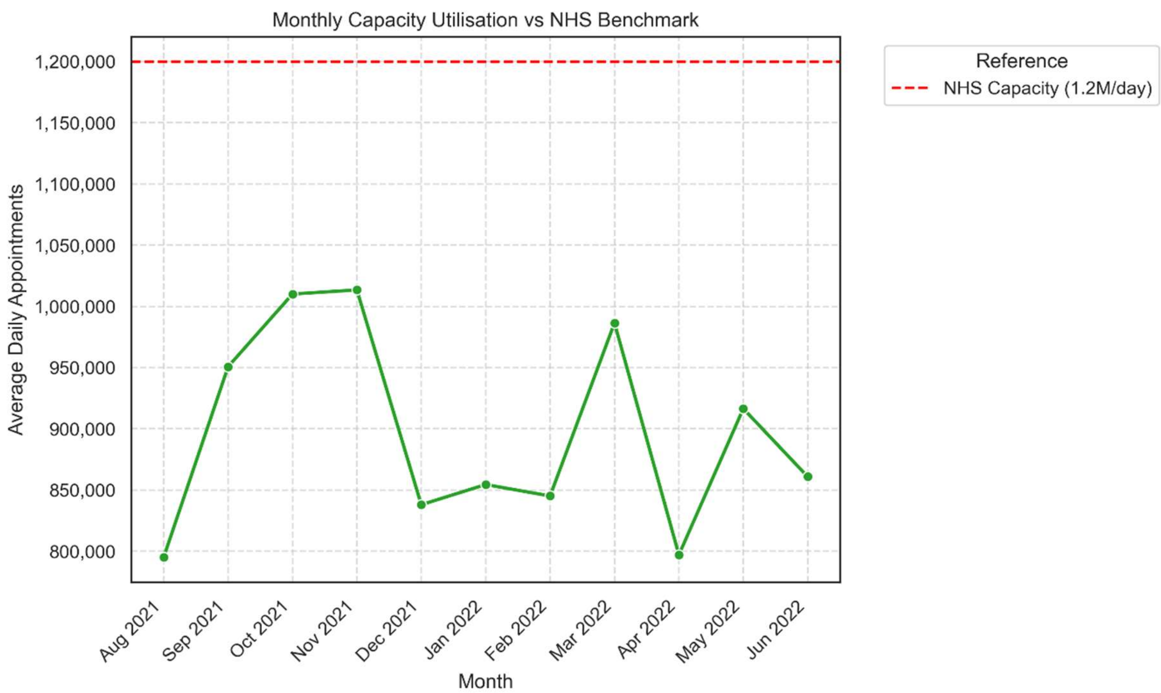

Figure 1

Insight: Average daily utilisation ranges between 800,000 and 1 million appointments against a 1.2 million capacity, indicating the NHS operates near 70–85% capacity with limited flexibility during peak demand.

Note: With the provided datasets, it was not possible to measure staffing adequacy. Results could be inferred indirectly if required but could be misleading.

2. *What was the actual utilisation of resources?*

Measure 1: Number of appointments per month by service setting, showing how utilisation varies across service types such as General Practice (GP), Primary Care Networks (PCNs), and community settings.

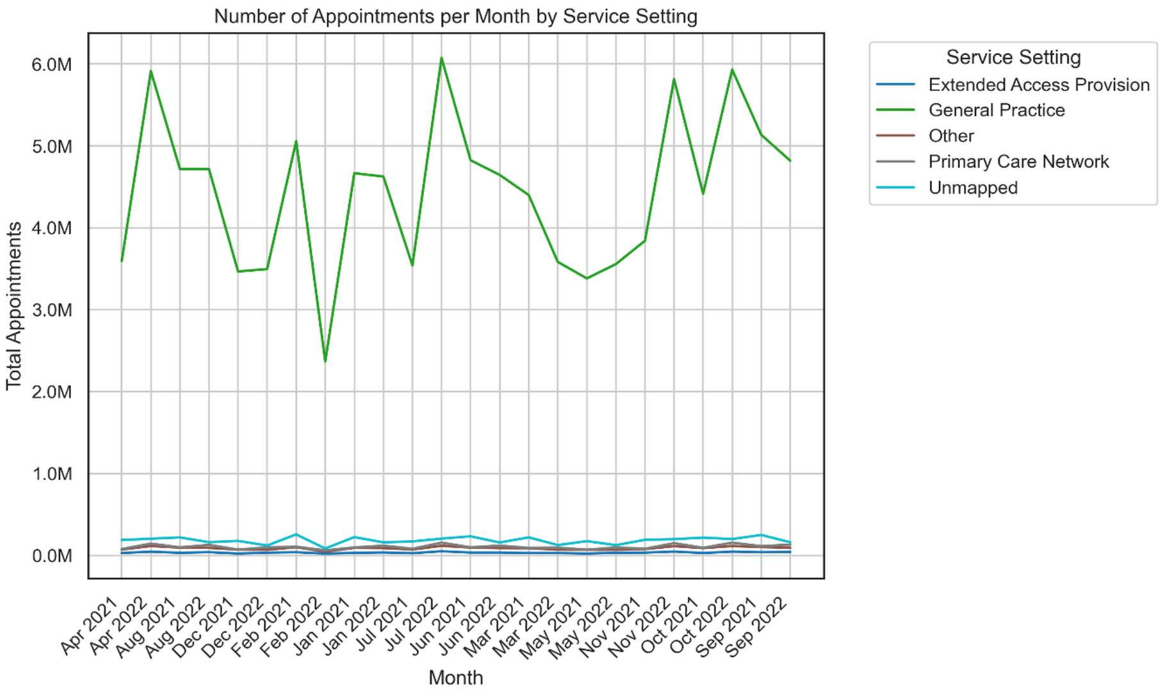

Figure 2

Insight 1: GP appointments dominate, ranging from 2.5 to 6 million per month—far exceeding other settings. The variation suggests changing demand or possible capacity limits, warranting further analysis.

Measure 2: Appointment volume by healthcare professional type, showing utilisation by staff category.

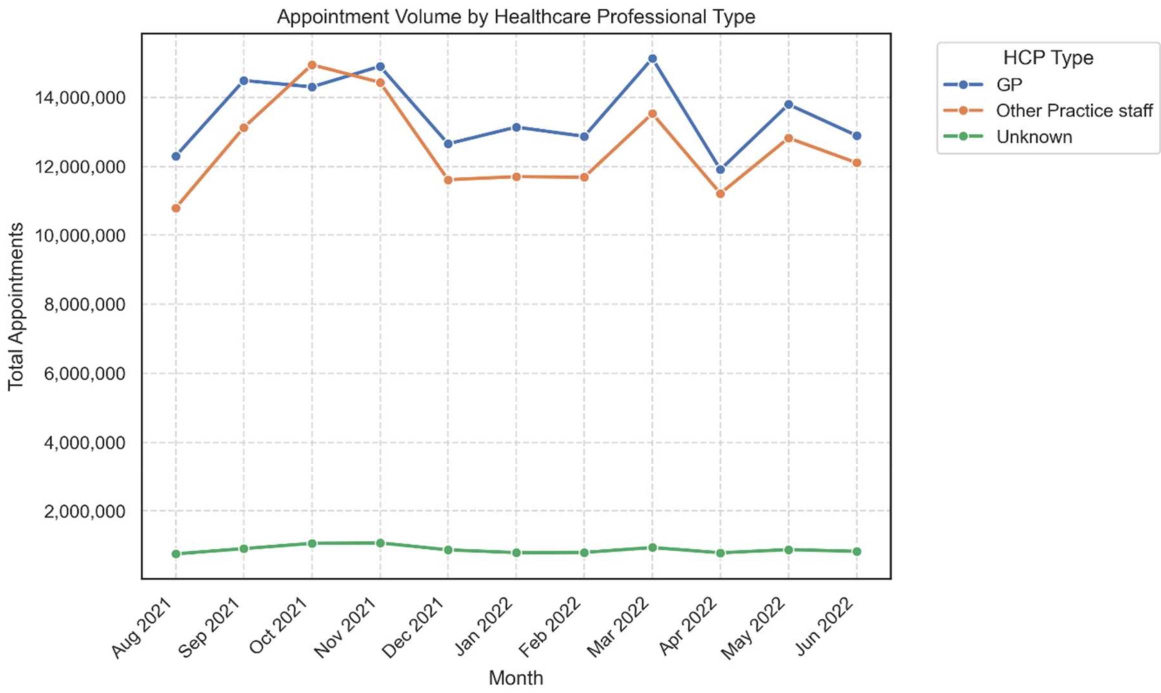

Figure 3

Insight 2: GP and other staff appointments stayed closely aligned at 11–14 million per month, indicating balanced demand across professional groups.

Measure 3: Number of appointments per month by national category. Shows the distribution of consultation types across the NHS.

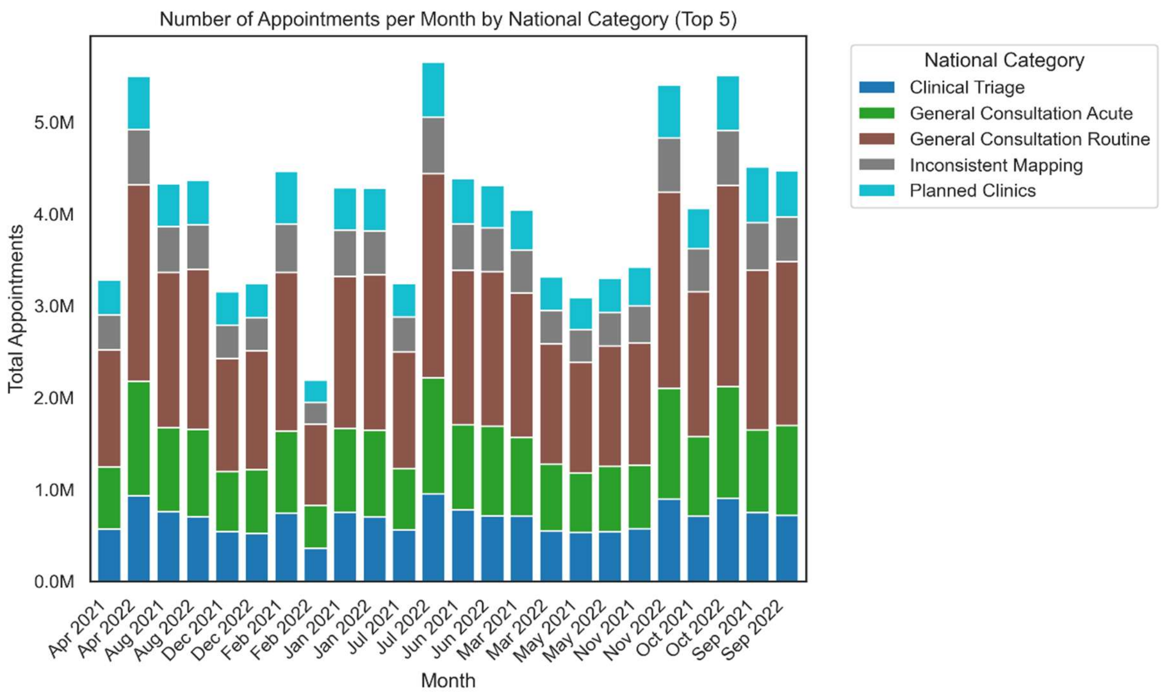

Figure 4

Insight 3: General Consultations dominate, while Clinical Triage and Planned Clinics form a smaller share, confirming GP consultations as the main driver of NHS appointment demand.

3. *What percentage of appointments were missed and how can this be reduced?*

Measure: Monthly appointment attendance, showing the rate of missed appointments.

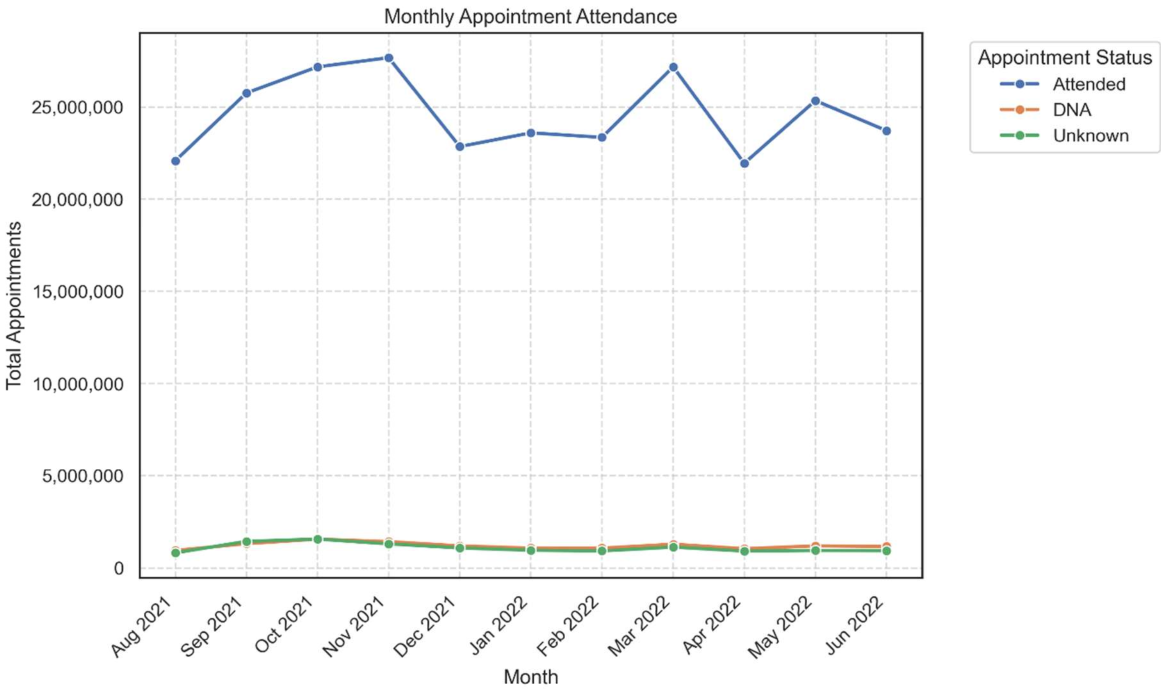

Figure 5

Insight: Around 1 million appointments are missed each month, indicating persistent inefficiencies. Improved reminders, easy cancellations, and shorter booking windows could help reduce this cost.

4. *Can social media be leveraged to provide indications of operational pressure?*

Measure: Hashtag and keyword usage in NHS-related tweets.

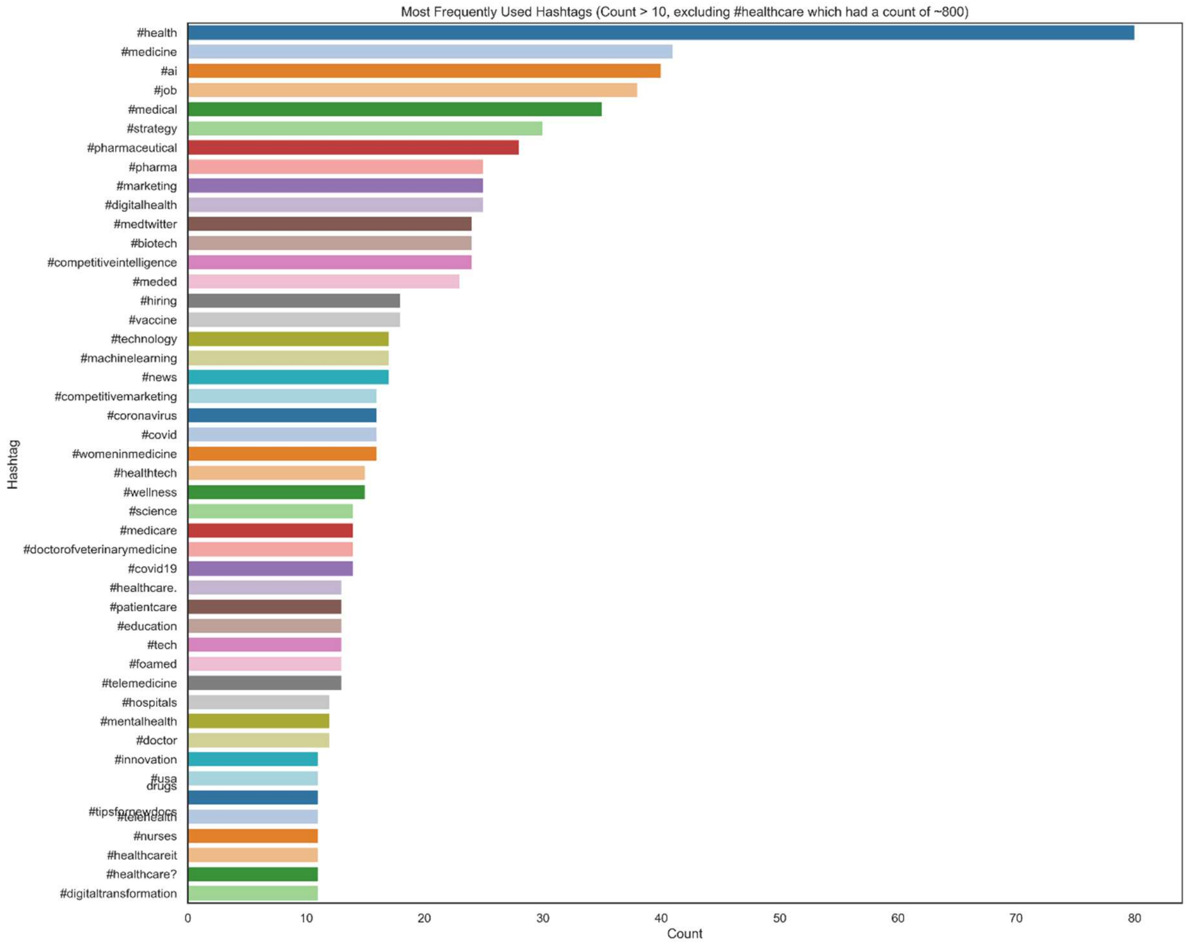

Figure 6 – This visual is illustrative showing how social media data could be integrated alongside operational metrics in future.

Insight: Hashtag and keyword trends reveal public sentiment and can indicate early signs of NHS capacity or service pressure.

Patterns and Predictions

General Practices

General Practice appointments remain the most popular across all service settings, with consistently high utilisation against the NHS's 1.2 million daily capacity benchmark. This sustained demand indicates that GP services are operating close to capacity, leaving limited flexibility to absorb further increases in patient volume. A revision of the benchmark may be useful. Additionally, targeted investment in high-demand areas—through additional staffing, extended hours, or efficiency measures—will be essential to maintain service volume and quality.

Missed appointments

Missed appointments remain steady at around one million per month, reflecting a persistent inefficiency across the system. Without targeted action, missed appointment rates are likely to remain stable, continuing to waste time and resources. Stronger reminder systems, simplified cancellations, and greater use of alternative or remote appointment modes could help reduce missed appointments and improve overall capacity utilisation.

Social Media

Social media trends can highlight public concern and service pressures; online discussions could therefore serve as early-warning indicators of rising operational strain. With more targeted research, social data could be integrated into monitoring systems to identify emerging issues in real time. Additionally, coordinated NHS campaigns could encourage the use of specific hashtags to improve tracking and support proactive communication with the public.

Conclusion

The analysis highlights sustained pressure within General Practice and persistent inefficiencies from missed appointments. Although current capacity exceeds demand, there is potential for limited headroom in future.

Consistent missed appointment levels signal a need for operational improvement. Strengthening data quality, enhancing appointment management, and exploring social media as a real-time insight source could support more agile, evidence-led NHS planning.